Course Code:BTCS 610-18	Course Title : Digital Image Processing	3L:0T:0P	3Credits

Detailed Contents:

UNIT 1: Introduction of Digital Image Processing (DIP)

Introduction to the DIP areas and applications; Components of Digital Image Processing; Elements of Visual Perception; Image Sensing and Acquisition; Image Sampling and Quantization; Relationships between pixels; color models.

[7hrs] (CO 1)

UNIT 2: Image Enhancement

Spatial Domain: Gray level transformations; Histogram processing; Basics of Spatial Filtering; Smoothing and Sharpening Spatial Filtering

Frequency Domain: Introduction to Fourier Transform; Smoothing and Sharpening frequency domain filters; Ideal, Butterworth and Gaussian filters

[10hrs] (CO 2)

UNIT 3: Image Restoration

Noise models; Mean Filters; Order Statistics; Adaptive filters; Band reject Filters; Band pass Filters; Notch Filters; Optimum Notch Filtering; Inverse Filtering; Wiener filtering

[8hrs] (CO 3)

UNIT4: Feature Extraction and Image Segmentation

Feature Extraction: Contour and shape dependent feature extraction, Extraction of textural features

Segmentation: Detection of Discontinuities; Edge Linking and Boundary detection; Region based segmentation; Morphological processing- erosion and dilation.

[10hrs] (CO 4)

UNIT 5: Image Compression and Encoding

Entropy-based schemes, Transform-based encoding, Predictive encoding and DPCM, Vector quantization, Huffman coding. **[10hrs](CO 5)**

Course Outcomes:

After undergoing this course, the students will be able to:

CO1: Understand the basic concepts of DIP.
CO2: Improve the quality of digital images.
CO3: Understand and De-noise Digital Images
CO4: Segment digital images and extract various features from digital images
CO5: Understand various image compression techniques and apply such techniques to compress digital images for reducing the sizes of digital images.

Suggested Readings/ Books:

1. Rafael C. Gonzales, Richard E. Woods, "Digital Image Processing", Third Edition, Pearson Education, 2010.
2. Anil Jain K. "Fundamentals of Digital Image Processing", PHI Learning Pvt. Ltd., 2011.
3. Willliam K Pratt, "Digital Image Processing", John Willey, 2002.
4. Nick Efford, "Digital Image Processing a practical introduction using Java", Third Edition, Pearson Education, 2004.
5. R.C. Gonzalez, R.E. Woods, and S. L. Eddins "Digital Image Processing using MATLAB", Pearson Prentice-Hall, 2004.
6. Sandipan Dey, "Hands-On Image Processing with Python", Packt, 2018

Course Code:BTCS 611-18	Course Title: Digital Image Processing Lab	L:0;T:0; P:	Credits

Detailed List of Tasks:

1. WAP to draw Histogram of digital Image
2. WAP to enhance the quality of digital image using various gray level transformations.
3. WAP to enhance the quality of digital image using Average and median filters in spatial domain.
4. WAP to convert digital image from spatial domain to frequency domain.
5. Implement low pass filters in frequency domain for image enhancement.
6. Implement high pass filters in frequency domain for image enhancement.
7. Implement Optimum Notch Filtering for de-noising of digital image.
8. WAP to segment digital image using thresholding approach.
9. WAP to extract shape and texture based features from image.
10. WAP to compress digital image using entropy based approach.

Suggested Tools – MATLAB/Python/JAVA